

Local Recurrence After Primary Proton Beam Therapy in Uveal Melanoma: Risk Factors, Retreatment Approaches, and Outcome



IRA SEIBEL, DINO CORDINI, MATUS REHAK, ANNETTE HAGER, ALINE I. RIECHARDT, ALEXANDER BÖKER, JENS HEUFELDER, ANDREAS WEBER, JOHANNES GOLLRAD, ANGELA BESSERER, AND ANTONIA M. JOUSSEN

- **PURPOSE:** To evaluate the risk factors, recurrence rates, retreatments, and long-term patient outcomes following proton beam therapy for uveal melanoma.
- **DESIGN:** Retrospective interventional case series.
- **METHODS:** All patients treated with primary proton beam therapy for uveal melanoma at the oncology service at Charité-Berlin and Helmholtz-Zentrum-Berlin between May 1998 and December 2008 were reviewed for local recurrence. Of 982 patients, 982 eyes matched the inclusion criteria. The data were obtained from electronic health records, operative reports, discharge letters, and radiation planning. Comparisons of fundus photographs and ultrasound measurements were performed to assess the growth pattern of the tumor and to determine the success of retreatment, in the case that a globe-retaining therapy was undertaken.
- **RESULTS:** Of 982 patients, 35 patients (3.6%) developed local recurrence. The median follow-up was 60.7 months (6.0–170.4 months). Local control rate was 96.4% and the overall eye retention rate was 95.0% in this cohort. Local recurrence was correlated with a higher risk for metastasis and reduced survival. Largest tumor diameter was identified as the sole statistically significant risk factor for local recurrence ($P = .00001$). All globe-retaining retreatment approaches for local recurrence, including proton beam therapy, brachytherapy, and transpupillary thermotherapy used for recurrences at the tumor margins, showed good local tumor control and similar metastasis-free survivals.
- **CONCLUSIONS:** This study showed that each globe-retaining retreatment approach can result in satisfying local tumor control. In case of early detection of local recurrence, preservation of the globe can be warranted. Therefore, regularly performed follow-ups should be ensured. (Am J Ophthalmol 2015;160(4):628–636. © 2015 by Elsevier Inc. All rights reserved.)

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From the Augenklinik (I.S., D.C., M.R., A.H., A.I.R., A.Böker, J.H., A.W., A.M.J.) and Klinik für Radioonkologie und Strahlentherapie (J.G., A.Besserer), Charité Universitätsmedizin Berlin, Campus Benjamin Franklin, Berlin, Germany; and Berlin Protonen am Helmholtz-Zentrum Berlin für Materialien und Energie (D.C., J.H., A.W.), Charité Universitätsmedizin Berlin, Lise-Meitner-Campus, Berlin, Germany.

Inquiries to Ira Seibel, Department of Ophthalmology, Charité, University Medicine Berlin, Campus Benjamin Franklin, Hindenburgdamm 30, 12200 Berlin, Germany; e-mail: ira.seibel@charite.de

THE 3 OBJECTIVES OF UVEAL MELANOMA TREATMENT are saving the patient's life, preservation of the globe, and preservation of its function, in descending order. With the decision for globe-retaining treatment, local tumor control becomes greatly important. Today's most accepted globe-retaining therapies are brachytherapy with episcleral plaques (eg, ^{125}I or ^{106}Ru), teletherapy with heavy charged particles (proton beam and helium ion beam therapy), and CyberKnife radiotherapy. Transpupillary thermotherapy can be used for small choroidal melanoma.¹ All approaches bear more or less risk of local recurrence. Proton beam therapy showed local tumor recurrence rates ranging from 2% to 8.4% after 5 years.^{2–11} Risk factors for failure of local control after proton beam therapy include a large tumor diameter, small tumor volume, iris root involvement, and safety margins inferior to 1 mm.³ Recurrence may take the form of marginal, diffuse, distant, or extrascleral extension.³ Currently, enucleation is still the most used treatment for recurrent melanoma, although irradiation (eg, proton beam therapy) is a suitable option for retreatment.¹² As there is no standard for retreatment, the treatment approach depends on the extent and the location of the recurrence and remains a case-by-case decision. This study therefore concentrates on classifying the risk factors for local recurrences after proton beam therapy and the local and systemic outcome of different retreatment approaches.

METHODS

THE STUDY WAS APPROVED BY THE INSTITUTIONAL REVIEW board of the Charité-Universitätsmedizin Berlin, Berlin, Germany, and was in accordance with the tenets of the Declaration of Helsinki. All patients with initial diagnosis of choroidal or ciliary body melanoma that were treated with primary proton beam therapy at the Ocular Oncology Service at Charité Berlin and the Helmholtz Center Berlin between May 1998 and December 2008 were reviewed for local recurrence. In the framework of our treatment planning, the indication for using primary proton beam therapy is a centrally located tumor close to sensitive structures or a peripheral tumor beyond the arcades exceeding 6 mm.

The dose for proton beam therapy in our study was 60 cobalt gray equivalent in total (cobalt gray equivalent: 1 cobalt gray equivalent = 1.1 Gy, taking a radiobiological effectiveness of 1.1 into account), given in 4 fractions of 15 cobalt gray equivalent on 4 sequential days.¹³ Prior to irradiation, 2.5-mm-diameter tantalum clips were sutured to the sclera to demarcate the margins of the tumor. Three-dimensional modeling of the tumor (clinical target volume) was calculated based on localization measurements via ultrasound, funduscopy, and computed tomography scans. Treatment planning and dose calculation were performed with the treatment planning systems EYEPLAN version 1.2 and (since 2006) OCTOPUS version 4.4 (German Cancer Research Center, DKFZ and Hahn-Meitner-Institute, Berlin). The “clinical target volume” is extended by a standard 2.5 mm margin to compensate for patient motion, set-up errors, range uncertainties, and lateral dose build-up. During treatment, the head of the patient was immobilized by a combination of an individually molded face mask and a bite block to ensure the correct position. The patient had to focus on an external light source. In case of movement, the therapy could immediately be interrupted.

Color fundus photographs taken throughout the examinations (before proton beam therapy [montage-photograph] until last follow-up) were compared to assess the growth pattern of the tumor in combination with ultrasound measurements. Comparisons of fundus photographs and ultrasound measurements were also performed to determine the success of retreatment, in the case that a globe-retaining therapy was undertaken.

The local recurrences were divided into 4 groups, depending on their growth pattern: marginal, central, distant, and diffuse. Local recurrence was defined as marginal if the tumor showed only a marginal growth with an increased diameter, central if the tumor increased only in thickness (after an interval of 12 months, to exclude the expected swelling during the first months after proton beam treatment), diffuse if diameter and height showed tumor growth, and distant if local recurrence occurred distant from the primary tumor.

• **STATISTICS:** For statistical analysis, patients were divided into 2 groups, a local control group and a local recurrence after proton beam therapy group. Statistical analysis was performed using SPSS 20.0 (SPSS, Inc, Chicago, Illinois, USA). A Mann-Whitney *U* test, Fisher exact test, χ^2 test, and factorial analysis of variance (ANOVA) were used to assess statistical differences between groups. Interrelationships between variables were assessed by calculating correlation coefficients according to Spearman/Pearson. Stepwise models for multiple regression analysis with odds ratios were used to evaluate different independent risk factors of the event of local recurrence (1 dependent and multiple independent variables), while models for multivariate regression analysis were used to evaluate risk factors in the development of

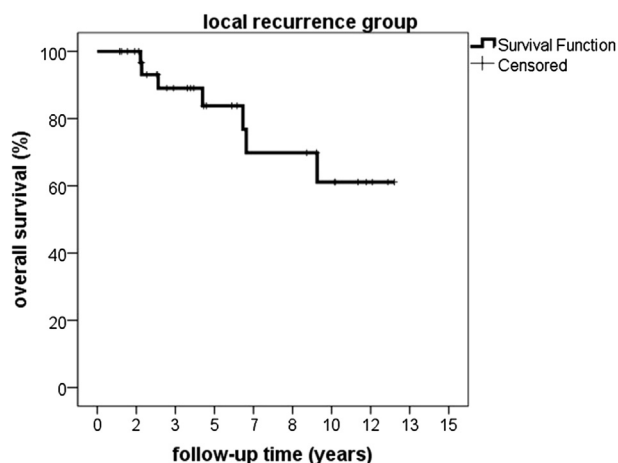


FIGURE 1. Overall survival in patients with local recurrence after proton beam therapy in uveal melanoma. Overall survival in % is displayed over years of follow-up in all patients who developed local recurrence after primary proton beam therapy in uveal melanoma, named local recurrence group. The Kaplan-Meier estimate for survival was 99% ($\pm 1.0\%$) at 1 year, 90% ($\pm 1.0\%$) at 3 years, 80% ($\pm 1.0\%$) at 5 years, and 60% ($\pm 1.0\%$) at 10 years in the local recurrence group.

local recurrences (multiple independent and dependent variables). Kaplan-Meier survival analysis was used for examining the distribution of time-to-event variables. Visual acuity is described as logMAR. In relation to the MARAN protocol, light perception (LP) was added as 2.1 logMAR, hand motion (HM) as 2.0 logMAR, and counting fingers (CF) as 1.9 logMAR.¹⁴

RESULTS

WE INCLUDED 982 PATIENTS, OF WHOM 35 PATIENTS (3.6%) developed local recurrence after proton beam therapy. The median follow-up was 60.7 months (6.0–170.4 months). The local tumor control rate was 96.4%. The Kaplan-Meier estimate for local recurrence was 2% ($\pm 1\%$) at 1 year, 3% ($\pm 1\%$) at 3 years, 4% ($\pm 1\%$) at 5 years, and 6% ($\pm 1\%$) at 10 years. After proton beam therapy, the median time local recurrence occurred was 17.5 months (1.3–88.7 months).

Time of overall survival differed significantly ($P < .05$) between 49.4 months (11.5–152.1 months) in the local recurrence group and 61.7 months (0.1–170.4 months) in the local control group. The Kaplan-Meier estimate for survival was 99% ($\pm 1\%$) at 1 year, 90% ($\pm 1\%$) at 3 years, 80% ($\pm 1\%$) at 5 years, and 60% ($\pm 1\%$) at 10 years in the local recurrence group compared to 99% ($\pm 1\%$) at 1 year, 95% ($\pm 1\%$) at 3 years, 89% ($\pm 1\%$) at 5 years, and 80% ($\pm 1\%$) at 10 years in the local control group (Figures 1 and 2).

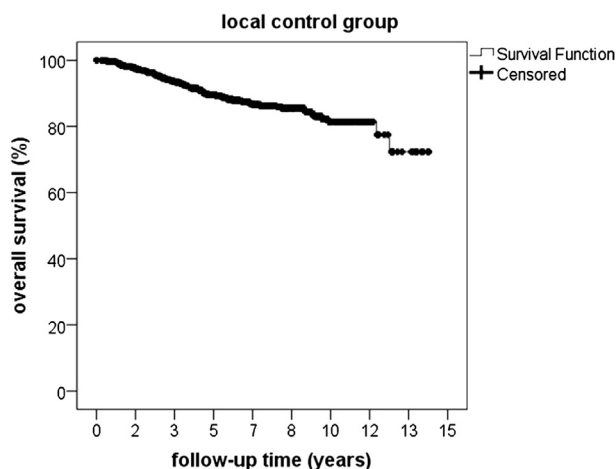


FIGURE 2. Overall survival in patients with local tumor control after proton beam therapy in uveal melanoma. Overall survival in % is displayed over years of follow-up in patients who showed local tumor control after primary proton beam therapy in uveal melanoma, named local control group. The Kaplan-Meier estimate was 99 % ($\pm 1.0\%$) at 1 year, 95% ($\pm 1.0\%$) at 3 years, 89% ($\pm 1.0\%$) at 5 years, and 80% ($\pm 1.0\%$) at 10 years in the local control group.

Metastasis occurred in 10 patients of the local recurrence group (10/35, 28.6%) after a median time of 53.9 months (12.3–147.9 months) after proton beam therapy and 14 months (2.6–128.6 months) after the development of local recurrence—compared to 145 patients (145/947, 15.3%) of the local control group who developed metastasis after a median interval of 31.6 months (0.4–154.1 months) after proton beam therapy, $P < .03$.

The risk of developing metastasis was statistically significantly increased in the local recurrence group, $P < .03$. Median time of metastasis-free survival was 49.4 months (11.5–148.8 months) and 60.7 months (6.0–170.4 months) in the local recurrence group and the local control group, respectively. The Kaplan-Meier estimate for metastasis-free survival calculated from the day after finishing proton beam therapy was 97% ($\pm 1\%$) at 1 year, 84% ($\pm 1\%$) at 3 years, 76% ($\pm 1\%$) at 5 years, and 50% ($\pm 1\%$) at 10 years in the local recurrence group; and 99% ($\pm 1\%$) at 1 year, 90% ($\pm 1\%$) at 3 years, 85% ($\pm 1\%$) at 5 years, and 77% ($\pm 1\%$) at 10 years in the local control group. Statistical analysis showed a significant difference between both groups, indicating a prolonged metastasis-free survival in patients without local recurrence, $P = .037$ (Figures 3 and 4).

• **EVALUATION OF RECURRENCE RISK FACTORS:** *Initial tumor volume.* Median initial tumor volume was 362 mm³ (31–2011 mm³) in the local recurrence group vs 276 mm³ (1.0–3063.0 mm³) in the local control group. No statistical significance was found between groups, nor was there any significant correlation with local recurrence.

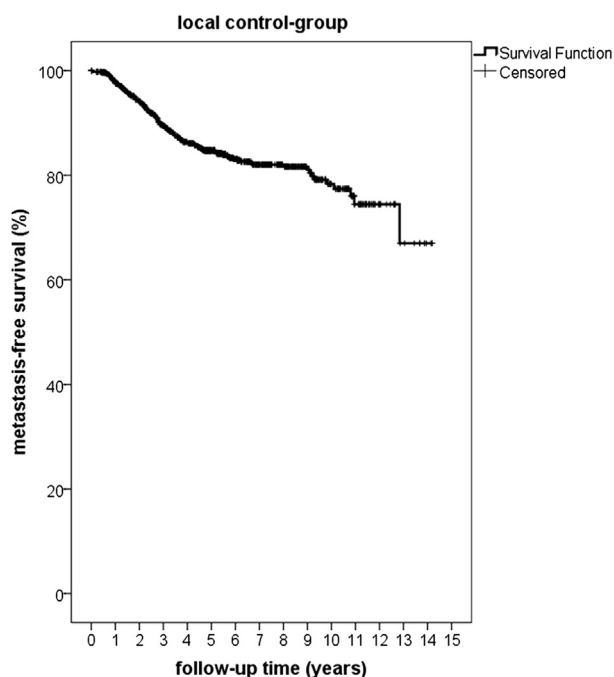


FIGURE 3. Metastasis-free survival in patients with local tumor control after proton beam therapy in uveal melanoma. Metastasis-free survival in % according to years of follow-up in all patients who showed local tumor control after primary proton beam therapy in uveal melanoma. The Kaplan-Meier estimate was 99% ($\pm 1.0\%$) at 1 year, 95% ($\pm 1.0\%$) at 3 years, 89% ($\pm 1.0\%$) at 5 years, and 80% ($\pm 1.0\%$) at 10 years in the local control group.

Initial tumor diameter. Median initial largest tumor diameters ranged between 12.5 mm (5.1–17.6 mm) and 12.0 mm (3.0–26.1 mm) in the local recurrence group and the local control group, respectively. Statistical univariate analysis revealed a strong correlation between tumor diameter and occurrence of local recurrence, $P < .0001$. Further evaluation of patients with large largest tumor diameter, defined as largest tumor diameter >10 mm, revealed that especially tumors with flat edges are at high risk for local recurrence owing to underestimation of clinical target volume.

Initial tumor thickness. Median initial tumor thickness was 5.1 (1.8–14.8 mm) in the local recurrence group and 4.4 mm (1.3–15.3 mm) in the local control group. No statistical significance was found between these groups, nor was there any significant correlation with local recurrence.

Distance to sensitive structures. Of 982 patients, 320 patients (32.6%) presented with a tumor directly adjacent to the optic disc or fovea, 320 (32.6%) had tumors at a distance to a sensitive structure of 0.1–2.5 mm, 176 (17.9%) at a distance of 2.6–5 mm, 119 (12.1%) at a distance of 5.1–10 mm, and 47 (4.8%) at a distance of 10.1–21 mm. All

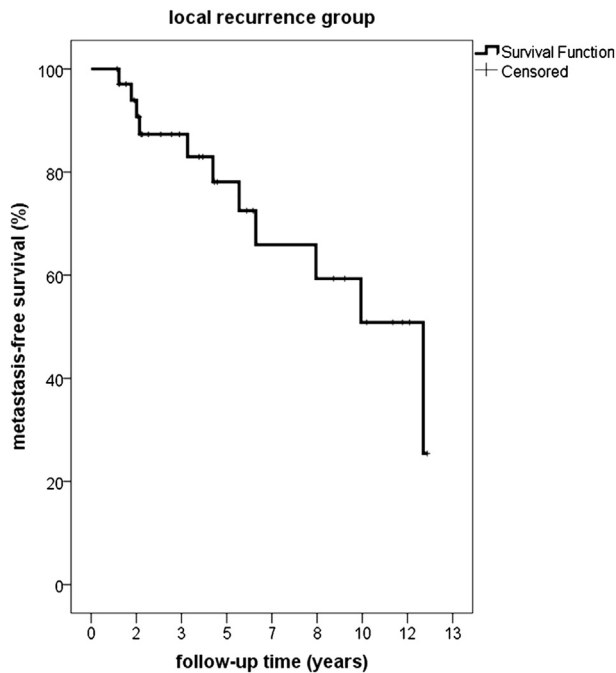


FIGURE 4. Metastasis-free survival in patients with local recurrence after proton beam therapy in uveal melanoma. Metastasis-free survival in % displayed over years of follow-up in all patients who developed local recurrence after primary proton beam therapy in uveal melanoma. The Kaplan-Meier estimate for metastasis-free survival was 97% ($\pm 1.0\%$) at 1 year, 84% ($\pm 1.0\%$) at 3 years, 76% ($\pm 1.0\%$) at 5 years, and 50% ($\pm 1.0\%$) at 10 years in the local recurrence group.

tumors that developed local recurrence were located at the posterior pole (31 patients) except those with ciliary body involvement (4 patients). The median distance was 0.5 mm (0–6.0 mm) to the optic disc and 0.4 mm (0–7.5 mm) to the fovea. In the local control group median distance to optic disc and fovea were 1.5 mm (0–21.0 mm) and 0.9 mm (0–18.9 mm), respectively. There was a statistically significant difference ($P < .005$) for distances to sensitive structures between both groups but no correlation related to the development of local recurrence (Table).

2007 Tumor-Node-Metastasis (TNM) classification. Of the 35 patients with local recurrence 5 (14.3%) were classified as T1, 14 (40%) as T2, 11 (31.4%) as T3, and 5 (14.3%) as T4. Of the 947 patients with good tumor control, 260 (27.4%) were classified as T1, 340 (35.9%) as T2, 278 (29.4%) as T3, and 69 (7.3%) as T4. There was no statistical significance found between groups, and no significant correlation with local recurrence.

Safety margin. The clinical target volume is expanded by a 1.0 mm safety margin in all directions. The resulting volume is called planning target volume. According to ICRU Reports 50 and 62, this margin should compensate

TABLE. Anatomic Features of All Patients That Were Treated With Primary Proton Beam Therapy for Uveal Melanoma According to Local Recurrence Group and Local Control Group

Anatomic Features	Initial Tumor Volume (mm ³)	Initial Tumor Diameter (mm)	Initial Tumor Thickness (mm)	Distance to Optic Disc (mm)	Distance to Fovea (mm)
Local recurrence group (n = 35 patients)					
Median	362	12.5	5.1	0.5	0.6
Min	32	5.1	1.8	0	0
Max	2011	17.6	14.8	6.0	7.5
Local control group (n = 947 patients)					
Median	276	12.0	4.4	1.5	1.9
Min	1	3.0	1.3	0	0
Max	3063	26.1	15.3	21.0	18.9

Max = maximum; Min = minimum.

for uncertainties in patient set-up, small movements of the treated eye during irradiation, and all other geometric uncertainties.^{15,16} In the direction of the proton beam a margin of 1.5 mm distal and proximal to the planning target volume is used to cover range uncertainties of the proton beam to cover uncertainties of the density of the patient's eye. To compensate for the lateral dose fall-off, a lateral margin of 1.5 mm between the patient collimator and the planning target volume is needed. All these margins combined result, by coincidence, in a 2.5 mm margin around the clinical target volume. The margin of the clinical target volume was ≥ 2.5 mm in all but 5 patients (5/35, 14.3%) in the local recurrence group vs 9 patients (9/947, 1.0%) in the local control group, $P < .005$. Of the 5 patients who developed local recurrence, 4 patients were treated with a reduced margin by 0.5 mm and 1 patient by 1.0 mm. In the local control group the safety margin was reduced by 0.5 mm in 2 patients, by 1.0 mm in 5 patients, and by 1.5 mm in 2 patients. Statistical univariate analysis showed a statistically significant correlation between the reduction of safety margin and the development of local recurrence, $P < .05$. Three patients 3/35, (8.5%) were treated with an enlarged safety margin in the local recurrence group and 24 patients (24/947, 2.5%) in the local control group for unknown tumor spread or unknown infiltration of the optic nerve. There was no statistical significance found between groups and no significant correlation with local recurrence.

Ciliary body involvement. Of all 982 patients, 80 patients presented with ciliary body involvement. Four of those patients (5%) developed local recurrence and were classified as being of the local recurrence group. Conversely, 76

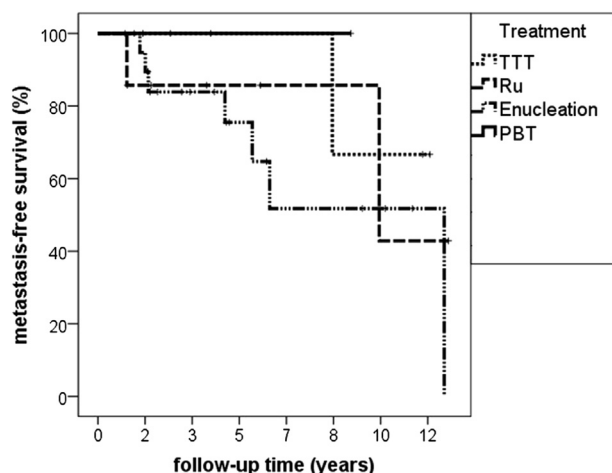


FIGURE 5. Retreatment approaches in patients with local recurrences occurring after proton beam therapy in uveal melanoma related to metastasis-free survival. Patients that developed local recurrence after primary proton beam therapy in uveal melanoma were treated with transpupillary thermotherapy (TTT), ruthenium brachytherapy (Ru), proton beam therapy (PBT), or enucleation for their local recurrences after primary proton beam therapy. This figure shows the metastasis-free survival in % according to years of follow-up dependent on retreatment approach.

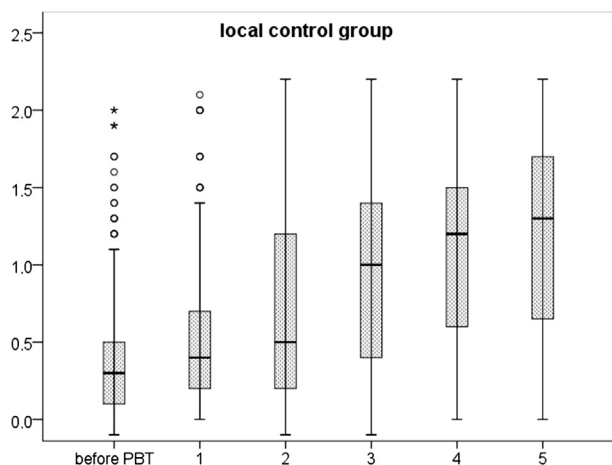


FIGURE 6. Visual outcome in patients with local tumor control after proton beam therapy in uveal melanoma. Visual acuity in all patients with local tumor control after primary proton beam therapy for uveal melanoma. Visual acuity (in logMAR) is displayed on the y-axis while years of follow-up are shown on the x-axis. PBT = proton beam therapy.

patients with ciliary body involvement (95%) showed good local tumor control. No statistical significance was found between these groups, nor was there any significant correlation with local recurrence.

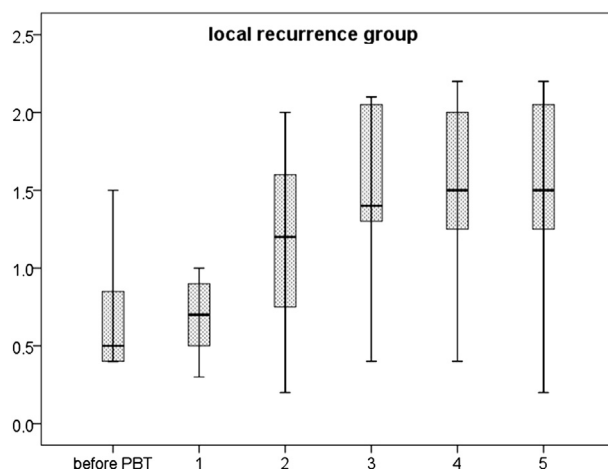


FIGURE 7. Visual outcome in patients with local recurrence after proton beam therapy in uveal melanoma. Visual acuity according to years of follow-up in all patients in whom local recurrence occurred after primary proton beam therapy for uveal melanoma. Visual acuity (in logMAR) is displayed on the y-axis while years of follow-up are shown on the x-axis. PBT = proton beam therapy.

• **MULTIVARIATE REGRESSION ANALYSIS:** Statistical multivariate regression analysis of the recurrence risk factors identified the initial largest tumor diameter as the sole statistically significant recurrence risk factor, $P = .00001$. All other features such as safety margin, close distances or involvement of papilla and fovea, initial tumor thickness, initial tumor volume, or ciliary body involvement were statistically unrelated to recurrence.

• **TREATMENT APPROACHES FOR LOCAL RECURRENCE:** Treatment of the 35 eyes with local recurrence comprised enucleation in 19 eyes (54%), brachytherapy in 8 eyes (23%), transpupillary thermotherapy in 7 eyes (20%), and proton beam therapy in 1 eye (3%). The treatment approach was based on an individual case-by-case decision. Transpupillary thermotherapy was only used for local recurrence at the tumor margins. No re-recurrence was detected. There was no statistically significant difference in metastasis-free and overall survival dependent on globe-retaining retreatment approaches but a reduced metastasis-free survival in patients treated with enucleation (Figure 5).

• **TYPES OF RECURRENCES:** The 35 local recurrences were divided into 4 groups depending on their growth pattern: 15 (42.9%) were classified as marginal, 9 (25.7%) as central, 4 (11.4%) as distant, and 7 (20%) as diffuse. There was no significant difference related to local tumor control, retreatment decision, metastasis-free survival, and overall survival between these 4 groups.

Three patients presented with recurrences following the distant growth pattern, with small pigmented lesions separate from the tumor's initial presentation.

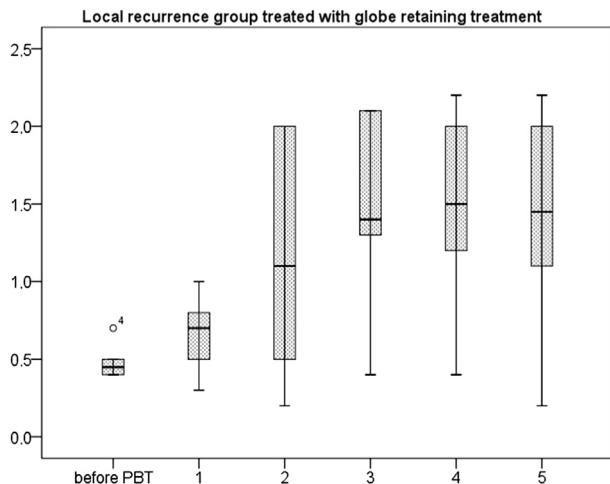


FIGURE 8. Visual outcome in patients who developed local recurrence after proton beam therapy in uveal melanoma and were treated with globe-retaining retreatment approaches for local recurrence. Visual acuity according to years of follow-up in patients who developed local recurrence after primary proton beam therapy in uveal melanoma and were retreated with globe-retaining treatment. Visual acuity (in logMAR) is displayed on the y-axis while years of follow-up are shown on the x-axis. PBT = proton beam therapy.

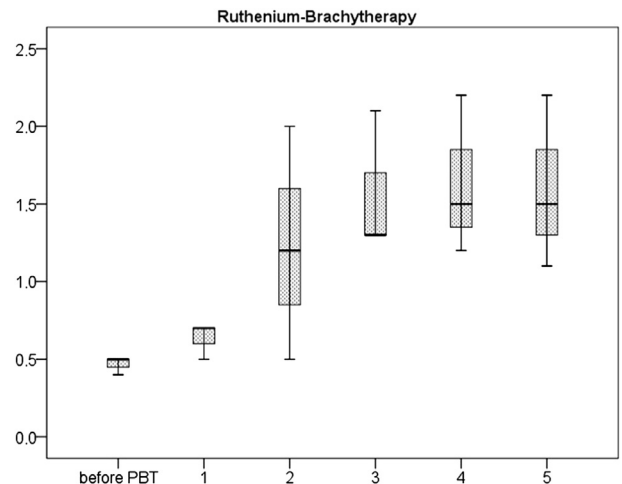


FIGURE 9. Visual outcome in patients in whom local recurrence occurred after proton beam therapy in uveal melanoma and who were treated with ruthenium brachytherapy for local recurrence. Visual acuity according to years of follow-up in patients who developed local recurrence after primary proton beam therapy in uveal melanoma and were retreated with ruthenium brachytherapy. Visual acuity (in logMAR) is displayed on the y-axis while years of follow-up are shown on the x-axis. PBT = proton beam therapy. (The median time of local recurrence was 13.5 months.)

These lesions later showed tumor growth and all 3 eyes were enucleated. Histologic examination revealed active tumor cells of malignant melanoma. The fourth patient, classified as distant recurrence, showed very flat elevations of the choroid in the ultrasound between primary tumor and local recurrence. These elevations were suspicious for ring melanoma but could not be conclusively clarified. This patient was successfully treated with brachytherapy.

• **FUNCTIONAL OUTCOME:** Median visual acuity decreased from 0.4 logMAR (0–1.5 logMAR) and 0.3 logMAR (0–2.9 logMAR) before proton beam therapy to 1.5 logMAR (0.4–2.0 logMAR) and 1.3 logMAR (0–2.2 logMAR) at last follow-up in the local recurrence and local control group, respectively. In the local recurrence group visual acuity decreased 2 years after proton beam therapy, the same year local recurrence on average occurred, compared to the local control group, in which vision dropped only after 3 years (Figures 6–8). If the local recurrence group is divided into the different retreatment approaches, there was no statistical difference between them in visual outcome (Figures 9 and 10). (The 1 patient treated with proton beam therapy could not take part in statistical analysis.) Statistical analysis revealed a strong correlation between the drop in visual acuity and the interval until retreatment ($P < .001$). Patients requiring retreatment

for local recurrence comprised 8 patients treated with brachytherapy for local recurrence after 13.5 months (1.3–58.8 months), 7 patients with transpupillary thermotherapy for local recurrence after 21.0 months (5.1–49.7 months), and 1 patient with proton beam therapy for local recurrence after 88.7 months (Figures 9 and 10).

• **COMPLICATIONS:** *Local control group.* Of the 947 patients, 115 (12.1%) developed neovascular glaucoma owing to radiation retinopathy after an average of 27.8 months (standard deviation 22.8 months), of whom 36 patients required cyclophotocoagulation or cyclocryocoagulation. Nevertheless, 30 of these patients had to be enucleated for neovascular glaucoma after 28.7 months (standard deviation 17.3 months). None of these globes had evidence of residual viable melanoma. Therefore the eye retention rate was 96.8% in patients without local recurrence, while the overall eye retention rate was 95.0%.

Local recurrence group. Of 35 patients who developed local recurrence, 7 patients (20%) suffered from neovascular glaucoma due to radiation retinopathy after an average of 25.1 months (standard deviation 18.9 months). Three patients (42.8%) underwent cyclophotocoagulation or cyclocryocoagulation. No patient required enucleation for neovascular glaucoma.

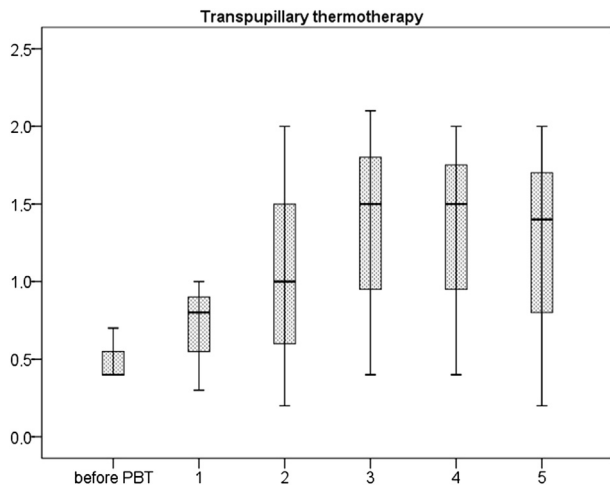


FIGURE 10. Visual outcome in patients with local recurrence occurring after proton beam therapy in uveal melanoma treated with transpupillary thermotherapy for the local recurrence. Visual acuity according to years of follow-up in patients who developed local recurrence after primary proton beam therapy in uveal melanoma and were retreated with transpupillary thermotherapy. Visual acuity (in logMAR) is displayed on the y-axis while years of follow-up are shown on the x-axis. PBT = proton beam therapy. (The median time of local recurrence was 21.0 months.)

DISCUSSION

PROTON BEAM THERAPY IS A SAFE TREATMENT OPTION FOR uveal melanoma, resulting in good local control and eye retention rates. Our local control rate of 96% and 94% at 5 and 10 years, respectively, was very similar to the rate described by Egger and associates (96% and 95%).⁴ Our local recurrence rate of 4% at 5 years is also similar to rates previously described (2%–8.4% at 5 years).^{1,3–11} The last recurrence in our study was detected 7.5 years after proton beam therapy, although later recurrences were described in other series, the latest occurring 16 years after local tumor control.^{1,3,4}

Marginal recurrences were the most frequently detected recurrences in our study. These results are consistent with other studies.^{1–3,5,8,12} We did not find any advantage of marginal recurrence over the other recurrence growth types, in particular with regard to survival prognosis, as mentioned by Caujolle and associates and Harbour and associates.^{3,17} In all our cases the reason for those recurrences was an underestimation of the clinical target volume. We did not find any errors in the irradiation planning performed by medical physicists by the use of wedges to spare sensitive structures or by wrong tantalum-clip location. These results, found by reviewing of records, are supported by univariate analysis that identified a reduced safety margin as a risk factor. This is in accordance with other studies.^{3,4} Even if not identified as a

recurrence risk factor in multivariate analysis, the safety margins are to be remembered as an important risk factor. Recommendations relating to how much reduction can be tolerated differ. Egger and associates⁴ detected a proportional association between the reduction of safety margin and the development of local recurrences, while Caujolle and associates³ identified a safety margin lower than 1 mm as a recurrence risk factor. Therefore, we recommend not reducing the safety margin, even in specific cases.

Three of the patients with distant recurrence growth pattern showed pigmented lesions surrounding the tumor. These lesions were not included in the clinical target volume, which may have appeared to be a mistake on subsequent examinations (ex post). Clinically, the differentiation between harmless freckles or nevi compared to flat tumor seeds can be challenging. In our center, treatment planning based on the intraoperatively (clip surgery) defined clinical target volume by the ocular oncologist is discussed and decided in an interdisciplinary conference consisting of radiation therapists, medical physicists, and ocular oncologists. This study emphasizes again the importance of the correct definition of the clinical target volume is a key factor for local tumor control.

Local recurrence was correlated with a higher risk for metastasis and a reduced survival time, as described by earlier studies.^{1,3–5,11,17} Apart from recurrence rate, tumor size, and tumor diameter, metastasis is associated with genetic risk factors. Our retrospective analysis does not involve assessment of genetic risk factors from the tumors.^{18,19} Therefore it remains unknown if local recurrence is an independent risk factor leading to metastasis or, rather, a consequence of the histologic and/or genomic aggressiveness of the initial tumor that is also responsible for the development of metastasis. If the cell division process of tumor cells is not finally inactivated by the chosen treatment, viable tumor cells may no longer be able to metastasize. Future studies that include genetic testing will be able to reveal more information, although up to the final clarification of this question local tumor control failure should be kept in mind as a potential reason for patients' reduction in life expectancy owing to metastasis.

In our study, multivariate analysis identified the largest initial tumor diameter as the sole local recurrence risk factor, which was also detected by other authors.^{1,3–6,8,10} Coujolle and associates assumed “that large and thin melanomas are more likely to recur because of the difficulty in defining the lateral tumoral limits properly,” to which we completely agree.³ It is impossible to detect the correct extent of a flat tumor by ultrasound. Sometimes even photographs hide tumor tissue owing to under- or overexposure, or owing to transpupillary illumination that is mostly used in conventional fundus cameras. The problem with transpupillary illumination is that it “is mostly reflected by the retinal pigment epithelium and inner choroid, so that tapering tumor edges are obscured.

Transscleral illumination allows more accurate definition of tumor extent, because it provides a better contrast between tumor margins and surrounding tissues.²⁰

This study showed that there was no difference between transpupillary thermotherapy, ruthenium brachytherapy, or proton beam therapy in long-term outcome for local recurrent melanoma, although conceding that only 1 patient was retreated with proton beam therapy. Enucleation was associated with a shorter metastasis-free survival. This is probably due to the fact that mainly eyes with advanced stages of local recurrences and poor prognosis were enucleated. No patient underwent secondary enucleation owing to retreatment failure after transpupillary thermotherapy, brachytherapy, or proton beam therapy. Riechardt and associates recently published nearly equal local control rates after salvage proton beam therapy compared to local control rates after primary proton beam treatment.¹² Marucci and associates showed that patients who underwent re-proton

beam therapy had a reasonable expectation of eye retention without any compromised survival, but their local control rate was reduced to 31% in patients retreated with proton beam therapy for local recurrences, suggestive of elevated radioresistance.^{21,22} In the present study, recurrences classified as central could give indication about an elevated radioresistance, but we revealed no re-recurrence after treatment. Furthermore genetic testing was not performed, allowing conclusion about radiosensitivity in our patients that developed local recurrences.²³

In conclusion, this study showed that each globe-retaining retreatment approach can result in satisfying local tumor control. Therefore a patient-specific therapy plan extensively discussed with the patient should be encouraged. Furthermore, we want to emphasize the importance of regularly performed follow-ups in an oncology service. Thus in case of early detection of local recurrence, preservation of the globe can be warranted.

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Confusing Eponyms

Alfred Hugh Benson, a surgeon at St. Mark's Ophthalmic Hospital in Dublin, Ireland, reported “A Case of Monocular Asteroid Hyalitis.” (*Trans Ophth Soc UK* 1894;14:101–104). James Moores Ball of St. Louis, Missouri described a similar case in 1924 and called it Benson disease, which is still a recognized but rarely used eponym. This condition is now called asteroid hyalosis. Ophthalmology also has Benson syndrome, which was described by D. Frank Benson, MD, a pioneer in the field of behavioral

neurology. (*Arch Neurol* 1988;45:789–793). The proper medical name of this syndrome is posterior cortical atrophy, a neurodegenerative disease of the visual cortex. The present trend in medical terminology is to avoid eponyms for several reasons, one of which is illustrated here in that the same eponym may apply to different diseases. In eponyms which are still used, the possessive use is now discouraged, since the author neither had nor owned the disorder. There is no known relationship between the two Benson physicians.

Robert M. Feibel, MD, Department of Ophthalmology and Visual Sciences, and Center for History Of Medicine, Washington University School of Medicine, St. Louis, MO. For the Cogan Ophthalmic History Society.